FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 1.3: TEMPERATURE AND THERMAL EXPANSION	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

AUDIENCE: This document is for STUDENTS to use as a model for answering the driving question.

DRIVING QUESTION: Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?

WHAT IS A FINAL EXPLANATION?

A Final Explanation is your opportunity to show deep understanding by explaining a real-world phenomenon using scientific concepts, evidence, and reasoning. Unlike a test that asks you to recall facts, a Final Explanation asks you to:

- Connect scientific concepts to an observable phenomenon
- Explain the mechanisms behind what you observe
- Apply scientific principles to solve real-world problems
- Communicate your understanding clearly and logically

Think of it as telling the complete story of why and how something happens, using the language and tools of physics.

ANCHORING PHENOMENON — THE CRACKED FROZEN BOTTLE:

A learner fills a clean 500 ml glass bottle completely to the brim with water, seals it tightly with a metal cap, and places it in a freezer overnight. The next morning the bottle is cracked — in some cases shattered — and a bulging column of ice protrudes where the cap once sat. This is puzzling because everyday experience tells us that things shrink when they get cold, yet the water clearly expanded enough to crack solid glass. Keep returning to this phenomenon throughout your Final Explanation.

INSTRUCTIONS FOR STUDENTS:

You will write a comprehensive explanation that addresses all five sections below. Your explanation should:

- Start with and return to the phenomenon: Refer back to the cracked frozen bottle throughout your explanation — it is the anchor for everything you write.
- Use scientific vocabulary accurately: temperature, thermal expansion, kinetic energy, anomalous expansion, coefficient of linear/volumetric expansivity, pressure, particle spacing, lattice structure.
- Include at least ONE calculation showing all working steps clearly.

- Draw connections between particle behaviour and observable, macroscopic changes.
- Apply concepts to real-world engineering solutions, using Kenyan examples where possible.
- Use headings, sub-headings, and diagrams where helpful.
- Write in a logical order so that each section builds on the previous one.

LENGTH GUIDELINE: 1,200–1,800 words (approximately 3–5 handwritten pages or 2–3 typed pages).
TIME ALLOCATION: You will have 2–3 class periods to complete this assessment, plus time for peer review and revision.

SCORING: Your work will be assessed using the rubric provided at the end of this document. Read the rubric carefully BEFORE you begin writing so you know exactly what excellence looks like.

Section 1: Temperature — What It Really Means at the Particle Level

Explain what temperature actually is, how it is measured, and why understanding it is essential to explaining the cracked bottle. **Guiding Questions:** – **What is temperature at the particle level? How does it relate to kinetic energy?** – **What are the two main temperature scales used in physics (Celsius and Kelvin), and how do you convert between them?** – **What types of thermometers exist, and how does each one work?** – **Which thermometer type would be most appropriate for measuring the temperature inside a freezer, and why?** – **What temperature change did the water in the bottle experience, going from room temperature (~25 °C) to a typical freezer temperature (~−18 °C)? Express your answer in**	Temperature is not just a number on a thermometer — it is a measure of the average kinetic energy of the particles (atoms or molecules) in a substance. When the water in the sealed bottle was placed in the freezer, its temperature dropped from approximately 25 °C to −18 °C. Converting to Kelvin: 25 °C + 273 = 298 K and −18 °C + 273 = 255 K. The temperature change was therefore $\Delta T = 298 - 255 = 43$ K (which is the same as 43 °C, since a change of 1 K equals a change of 1 °C). This 43-degree drop meant that every water molecule in the bottle was losing kinetic energy rapidly — slowing down, moving closer together at first, and eventually locking into a rigid crystal structure. To measure such low temperatures accurately, a thermocouple or a thermistor would be the most appropriate choice. A thermocouple works by joining two different metals (for example, copper and constantan) at a junction; when the junction experiences a temperature change, a small but measurable voltage is produced, which is converted to a temperature reading. Thermocouples can measure temperatures as low as −200 °C and are used in industrial freezers across Kenya's cold-chain food storage facilities, such as those used to preserve fish from Lake Victoria before transport to Nairobi markets. A thermistor (a semiconductor device) is equally useful for freezer applications because its electrical resistance changes dramatically with temperature, allowing very sensitive, digital readings. A liquid-in-glass thermometer (filled with alcohol, not mercury, since mercury freezes at −39 °C) would also work in a home freezer. The alcohol contracts as temperature falls, and the length of the alcohol column corresponds to the temperature on the calibrated scale — a direct application of thermal contraction in a liquid. Understanding temperature at this particle level is essential: the cracked bottle is not a mystery once you recognise that a 43 K drop in temperature did not simply slow the water molecules down uniformly — it triggered a phase change with a uniquely unexpected outcome.

both Celsius and Kelvin. **Key Concepts to Address:** – **Temperature as a measure of the average kinetic energy of particles — when temperature drops, particles move more slowly and lose kinetic energy.** – **The Celsius scale (based on water's freezing point at 0 °C and boiling point at 100 °C at standard pressure).** – **The Kelvin scale (absolute scale; 0 K = absolute zero = −273 °C). Conversion: K = °C + 273.** – **Thermometer types: liquid-in-glass (e.g., alcohol thermometer), bimetallic strip thermometer, thermocouple, resistance temperature detector (RTD), thermistor, infrared thermometer — explain the operating principle of at least THREE.** – **Identify the most suitable thermometer for a freezer context and justify your choice.** **Connection to Phenomenon: The water in the sealed bottle experienced a drop in temperature from roughly 25 °C (298 K) to −18 °C (255 K) — a change of 43 °C (or 43 K). Reducing the kinetic energy of water molecules is the starting point for the entire chain of events that cracked the bottle. Without understanding what temperature is and how to measure it precisely, we cannot**	

explain anything that follows.	

Section 2: Thermal Expansion and Contraction — The General Rule (and Its Famous Exception)	
Explain the general principle of thermal expansion for solids, liquids, and gases, and then explain why water is a famous exception when it freezes. **Guiding Questions:** – **What happens to the particles of a solid, liquid, or gas when temperature increases? Why does the substance expand?** – **What is the coefficient of linear expansivity (α) and how is it used in calculations? What is volumetric (cubical) expansivity, and how is it related to linear expansivity?** – **How do solids, liquids, and gases compare in the amount they expand for the same temperature change?** – **What is the 'anomalous expansion of water'? What happens to the density of water as it cools from 4 °C to 0 °C, and why?** – **How does the hexagonal ice crystal lattice explain why ice takes up MORE space than liquid water?** – **Use a calculation to estimate the volume increase when 500 ml of water freezes into ice (density of water ≈ 1,000 kg/m³; density of ice ≈ 917 kg/m³).**	The general rule in physics is this: when a substance is heated, its particles gain kinetic energy, vibrate more vigorously, and push each other further apart — so the substance expands. When cooled, the reverse happens: particles lose energy, attractions between them pull them slightly closer, and the substance contracts. This rule applies to solids, liquids, and gases, though gases expand far more dramatically than liquids, which in turn expand more than solids. For solids, we quantify this using the coefficient of linear expansivity, α (units: K^{-1}). The formula is: $\Delta L = \alpha \times L_0 \times \Delta T$ where ΔL is the change in length, L_0 is the original length, and ΔT is the temperature change. For volumetric (three-dimensional) expansion, the volumetric expansivity γ ≈ 3α, so: $\Delta V = \gamma \times V_0 \times \Delta T = 3\alpha \times V_0 \times \Delta T$ However, water breaks the general rule in a remarkable way — and this is the key to understanding the cracked bottle. Water reaches its maximum density at 4 °C. As it cools below 4 °C, instead of continuing to contract, it begins to EXPAND. By the time it freezes at 0 °C, its molecules have arranged themselves into a hexagonal (honeycomb-like) crystal lattice, held together by hydrogen bonds. In this lattice, each water molecule is bonded to four neighbours in a ring structure, and the rings contain large open spaces. This means ice is LESS dense than liquid water — which is why ice floats on the surface of Lake Victoria and why fish survive under a surface layer of ice in cold climates. Now apply this to the bottle. The bottle contained 500 ml of water. The mass of this water is: mass = density × volume = 1,000 kg/m³ × 0.000500 m³ = 0.500 kg When this water freezes completely into ice (density ≈ 917 kg/m³), the volume it occupies becomes: Volume of ice = mass ÷ density of ice = 0.500 kg ÷ 917 kg/m³ ≈ 0.000545 m³ = 545 ml The volume increased by approximately 45 ml — a 9% increase. But the rigid, sealed glass bottle could not accommodate this extra 45 ml. The expanding ice pushed outward with enormous force — estimates suggest pressures exceeding 25,000 kPa (about 250 times atmospheric pressure) can build up in a sealed container of freezing water. The glass, though hard, is brittle and cannot flex. It cracked along lines of least resistance, and the ice bulged upward through the cap — exactly what was observed the next morning.

Key Concepts to Address:

- General principle: when heated, particles gain kinetic energy → vibrate more vigorously → push further apart → substance expands. When cooled, the reverse occurs.
- Linear expansivity formula: $\Delta L = \alpha \times L_o \times \Delta T$, where ΔL = change in length, α = coefficient of linear expansivity (units: K^{-1} or $°C^{-1}$), L_o = original length, ΔT = temperature change.
- Volumetric expansivity $\gamma \approx 3\alpha$.
- Gases expand far more than liquids, which expand more than solids (for the same ΔT).
- Anomalous expansion of water: water is densest at 4 °C; below 4 °C it expands rather than contracts as it cools. At 0 °C, it freezes into a hexagonal lattice where each water molecule is hydrogen-bonded to four others in a ring structure, forcing the molecules further apart than in liquid water. This is why ice floats on water.
- Calculation: Volume of ice formed from 500 ml water = mass ÷ density of ice = (0.500 kg) ÷ (917 kg/m^3) ≈ 545 ml. Volume increase ≈ 45 ml — a 9% increase. This 9% volume increase, trapped inside a sealed rigid glass bottle, is what generated the enormous pressure that cracked the

glass. **Connection to Phenomenon: The cracked bottle is direct evidence of the anomalous expansion of water. If water behaved like most other liquids and simply contracted all the way down to 0 °C, the bottle would have been safe. Instead, below 4 °C, water does something extraordinary.**	

Section 3: Forces, Pressure, and Why the Glass Cracked	
Explain the relationship between thermal expansion, pressure, and the mechanical failure of the bottle. Connect particle-level behaviour to the macroscopic (large-scale) evidence you can observe. **Guiding Questions:** **– Why does expansion inside a confined space generate pressure?** **– How do the crack patterns on the bottle (running along the length of the bottle, with the ice plug protruding at the top) serve as evidence for the direction of the expansive force?** **– How is the situation in the bottle similar to natural geological phenomena such as frost weathering of rocks?** **– What properties of glass make it vulnerable to this kind of stress? What material would have survived? (Think about**	When the water in the sealed bottle began to freeze, it tried to expand by approximately 9% in volume. But the glass walls and metal cap prevented this expansion. Because volume cannot increase freely, the growing ice exerted an outward force on every surface it was in contact with. Using the relationship: Pressure = Force ÷ Area we can understand why this is so dangerous. The force is spread across the inner surface area of the bottle, but as more water freezes and the ice presses harder, the pressure rises rapidly — potentially to tens of thousands of kilopascals. Glass is a brittle material: it is hard and rigid, but it has very low tensile strength, meaning it cannot stretch or flex even slightly before it fractures. Once the pressure exceeded the tensile strength of the glass walls, the bottle cracked. The pattern of cracking is itself scientific evidence. The long, vertical cracks running up the sides of the bottle tell us that the main force was directed radially outward (pushing the walls apart). The protruding ice plug at the top tells us that the metal cap — the weakest and most flexible point — was pushed upward and outward as the last escape route for the expanding ice. If you observe the bottle cross-section diagram provided by your teacher, you can trace exactly how the hexagonal ice lattice grew and pushed in all directions simultaneously. This same process operates at a geological scale in the highland regions of Kenya. On the upper slopes of Mount Kenya and the Aberdare Ranges, water seeps into tiny cracks in exposed rock surfaces. At night, temperatures drop below 0 °C, the water freezes, expands, and widens the crack. After thousands of freeze-thaw cycles, massive boulders are split apart — a process called frost weathering. The physics of the cracked bottle and the split boulder are identical. Engineers must account for thermal expansion forces every day. Consider the Standard Gauge Railway (SGR) line running from Mombasa to Nairobi. Steel rails expand in the daytime heat of the Kenyan coast and contract at night. If rails were welded end-to-end with no gaps, the expansion force would buckle the tracks dangerously. Engineers therefore leave precisely calculated expansion joints between rail sections — small gaps that accommodate the expected $\Delta L = \alpha \times L_0 \times \Delta T$ change in length. Similarly, the Nyali Bridge in Mombasa and road bridges across the Rift Valley are built with expansion joints that allow the concrete and steel to move safely with temperature changes.

plastic water bottles — why do they bulge but not shatter?) – **How does this connect to engineering design? Give at least ONE Kenyan or African engineering example where thermal expansion forces must be accounted for.** **Key Concepts to Address:** – **Pressure = Force ÷ Area. When volume tries to increase inside a fixed container, force is exerted on the walls; because the area is fixed, pressure rises.** – **Stress and strain: brittle materials (like glass) fracture when stress exceeds their tensile strength; ductile/elastic materials (like plastic or copper) deform (stretch or bulge) without fracturing.** – **The crack pattern as observable evidence: longitudinal cracks running up the bottle walls show the ice pushed radially outward; the protruding ice plug shows upward force where the cap was the weakest point.** – **Frost weathering (physical weathering): water enters cracks in rocks, freezes, expands, and widens the crack — a geological process very relevant to highland Kenya (e.g., Mount Kenya, Aberdare ranges).** – **Engineering example: expansion joints in the**	Unlike glass, a plastic water bottle placed in the freezer does not shatter — it bulges and deforms. This is because plastic is a ductile (flexible) material with a higher tolerance for deformation before failure. Knowing the difference between brittle and ductile materials — and understanding thermal expansion forces — is therefore critical knowledge for every engineer, builder, and infrastructure designer in Kenya.

Nairobi–Mombasa Standard Gauge Railway (SGR) tracks, or expansion gaps in the Nyali Bridge in Mombasa, prevent structural damage from daily thermal expansion and contraction of steel. **Connection to Phenomenon:** The cracked bottle is a miniature version of the same force that splits boulders on Mount Kenya's peaks. The physics is identical — water expands by ~9% on freezing, generating pressure that brittle materials cannot withstand.	

Section 4: Does All Matter Behave the Same Way? Comparing Solids, Liquids, and Gases	
Expand the explanation beyond water to show your understanding of how ALL states of matter respond to temperature changes, and why the rates of expansion differ. **Guiding Questions:** – **How does a solid, a liquid, and a gas each respond when heated? Use the particle model to explain differences in the amount of expansion.** – **What is the difference between linear expansivity (used for solids) and volumetric/cubical expansivity (used for liquids and gases)? Why do we only talk about linear and superficial expansivity for solids?**	Not all matter expands at the same rate when temperature changes, and understanding these differences helps us see exactly why the water bottle cracked while a balloon filled with air would simply shrink in the freezer (and re-expand when warmed up again). SOLIDS expand the least for a given temperature change. In a solid, particles are held in fixed lattice positions and can only vibrate — they cannot move past each other. As temperature rises, vibrations become more energetic, and the average distance between particles increases very slightly. We quantify this using linear expansivity (α), because solids have a definite shape and we measure expansion along one dimension at a time. An everyday Kenyan example: the steel rails of the SGR line between Mombasa and Nairobi are installed with small expansion gaps. On a hot afternoon at the Mombasa terminus (temperature ~35 °C), a 25-metre steel rail (α for steel ≈ $12 \times 10^{-6}\ K^{-1}$) would expand by: $\Delta L = \alpha \times L_0 \times \Delta T = 12 \times 10^{-6} \times 25 \times 15 \approx 0.0045\ m = 4.5\ mm$ This may seem small, but across hundreds of kilometres of track, the cumulative expansion is substantial and must be engineered for. LIQUIDS expand more than solids for the same temperature rise, because the particles, while still close together, are not locked in fixed positions and can move more freely. We describe liquid expansion using volumetric (cubical) expansivity (γ). A practical Kenyan example: fuel sold at a petrol station in Nakuru is measured by volume. On a hot afternoon, the petrol in the underground storage tank has expanded compared to a cool morning — meaning the same mass of fuel occupies a larger volume. Fuel companies must account for this to ensure fair measurement. GASES expand dramatically — far more than solids or liquids — because their particles are already widely spaced and moving rapidly. At constant pressure, the volume of a gas is directly proportional to its absolute (Kelvin) temperature (Charles's Law: $V_1/T_1 = V_2/T_2$). A Kenyan example: a football inflated to the correct pressure in the cool highland town of Eldoret (altitude ~2,100 m, temperature ~15 °C = 288 K) and then taken to Mombasa (temperature ~35 °C = 308 K) will have a larger volume at the same

– Give a real-life Kenyan example of thermal expansion for EACH state of matter: (a) a solid, (b) a liquid, (c) a gas. **– Does all water expand when it freezes? What about salt water (like the Indian Ocean near Mombasa)? Briefly consider how dissolved salts affect the freezing point.** **Key Concepts to Address:** **– Solids: particles are in fixed positions, vibrating. Expansion is relatively small. We measure linear expansivity (α) and superficial (area) expansivity (β = 2α). Examples: steel rail gaps on SGR, overhead power lines sagging on hot Nairobi afternoons.** **– Liquids: particles can flow but are still close together. Expansion is greater than solids. Liquid-in-glass thermometers rely on this. Example: petrol/diesel in a fuel tank at Nakuru filling station expands on a hot afternoon — fuel is sold by volume, so temperature matters.** **– Gases: particles are far apart and move freely. Gases expand dramatically — the volume of a gas at constant pressure is directly proportional to its absolute temperature (Charles's Law: $V_1/T_1 = V_2/T_2$). Example: a football**	pressure: $V_2 = V_1 \times T_2/T_1 = V_1 \times 308/288 \approx 1.07 \times V_1$ The ball expands by about 7% — it may feel harder and could be at risk of over-pressure. Returning to water: most liquids contract continuously as they cool toward their freezing point. Water does this too — but only down to 4 °C. Below 4 °C, it reverses and expands as the hexagonal hydrogen-bonded lattice begins to form. This anomalous behaviour does not occur in saltwater in exactly the same way. Seawater near Mombasa (salinity ~35 g/L) has a freezing point of approximately −1.8 °C rather than 0 °C, because the dissolved sodium and chloride ions interfere with hydrogen bond formation, requiring lower temperatures before the ice lattice can assemble — a phenomenon called freezing point depression. This is why the Indian Ocean does not freeze even during the coldest Kenyan nights.

inflated in the cool highlands of Eldoret may become over-inflated when taken to the hot coast at Mombasa. – Anomalous expansion of water is unique among common substances — most liquids contract all the way to their freezing point. – Saltwater (like the Indian Ocean) has a lower freezing point than pure water (around −1.8 °C for typical seawater) because dissolved salt ions disrupt the formation of the hydrogen-bonded ice lattice — a phenomenon called freezing point depression. **Connection to Phenomenon: The cracked bottle reveals what is special about water compared to ALL other substances. Understanding the comparison helps us see why water's anomalous behaviour is so scientifically significant and so practically important — for life on Earth, for Kenyan lakes, and for engineering.**	

Section 5: Bringing It All Together — Your Complete Explanation and Real-World Applications	
In this final section, write a cohesive, complete explanation that directly answers the driving question. Then connect what you have learned to at least TWO real-world applications or engineering solutions.	COMPLETE ANSWER TO THE DRIVING QUESTION: The sealed glass bottle cracked in the freezer because of the anomalous expansion of water. When the temperature of the water dropped from approximately 25 °C (298 K) to −18 °C (255 K) — a change of 43 K — the water molecules lost kinetic energy and slowed down. Like most matter, water first contracted as it cooled. However, below 4 °C, water begins to behave differently from almost every other known liquid: instead of continuing to contract, it expands as its molecules arrange themselves into a hexagonal hydrogen-bonded ice crystal lattice. This lattice has large open spaces between molecules, making ice approximately 9% less dense than liquid water.

Guiding Questions:

- **In your own words, write a complete, concise answer to the driving question: 'Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?'**
- **What would happen if water did NOT have this anomalous expansion property? (Think about Lake Victoria, aquatic life in cold climates, and life on Earth generally.)**
- **Describe at least TWO real-world applications or engineering solutions that make use of — or must protect against — thermal expansion. Use Kenyan or African examples where possible.**
- **What is ONE new question you now have that this investigation has not fully answered? (Add this to your Driving Question Board.)**

Key Concepts to Address:

- **Synthesise: temperature → particle kinetic energy → phase change → anomalous expansion of water → volume increase of ~9% → pressure on glass walls → brittle fracture.**
- **Ecological importance of water's anomalous expansion: ice forms on the surface of water bodies (because ice is less dense), insulating the water below**

Inside the sealed, rigid glass bottle, this 9% volume increase had nowhere to go. The expanding ice exerted tremendous outward pressure on the glass walls (estimated pressures can exceed 25,000 kPa). Glass is a brittle material — it is rigid and hard, but it cannot bend or stretch. When the pressure exceeded the tensile strength of the glass, the bottle fractured along vertical lines where stress was greatest, and the ice plug was forced upward through the cap — the weakest point — exactly as observed.

This phenomenon reveals something profound about all matter: every substance changes its volume in response to changes in temperature. For most solids, liquids, and gases, cooling causes contraction. But water's anomalous expansion below 4 °C makes it a unique and extraordinary substance — and this uniqueness has consequences far beyond a cracked bottle.

WHAT IF WATER DID NOT EXPAND WHEN IT FROZE?

If water contracted all the way to its freezing point like most liquids, ice would be denser than liquid water — and it would sink. In Lake Victoria and other water bodies, ice would accumulate at the bottom, freezing the lake from the bottom upward, making aquatic life in cold seasons nearly impossible. Instead, because ice floats, it forms a surface layer that insulates the warmer water below. Fish, plants, and microorganisms survive beneath this icy lid. Water's anomalous expansion is not a curiosity — it is one of the reasons life on Earth can exist in cold climates.

REAL-WORLD APPLICATIONS:

1. Bimetallic Strips in Fire Alarms and Thermostats:

Two metals — for example, brass ($\alpha \approx 19 \times 10^{-6}$ K^{-1}) and steel ($\alpha \approx 12 \times 10^{-6}$ K^{-1}) — are bonded firmly together. Because brass expands more than steel for the same temperature rise, the strip bends toward the steel side when heated. In a fire alarm, this bending closes an electrical circuit and triggers the alarm. Schools and hospitals across Kenya increasingly rely on bimetallic-strip fire detection systems. The device works entirely because of the difference in linear expansivity between two metals — thermal expansion, used purposefully.

2. Expansion Joints in the SGR and Kenya's Road Bridges:

The Standard Gauge Railway between Mombasa and Nairobi passes through regions with large daily temperature swings — from the hot, humid coast to the cooler highlands. Engineers calculated the expected thermal expansion of steel rails using $\Delta L = \alpha \times L_0 \times \Delta T$, and designed expansion joints (small deliberate gaps) at regular intervals. Without these joints, the rails would buckle in daytime heat and crack under night-time contraction — exactly as our bottle cracked, but on a much larger scale. The same principle governs the design of concrete road surfaces across the Rift Valley, where expansion gaps prevent dangerous buckling.

A NEW QUESTION FOR THE DRIVING QUESTION BOARD:

If engineers can calculate exactly how much water expands when it freezes, could they design a container — using a specific material, shape, or thickness — that could safely contain freezing water without cracking? What material would they choose, and what would the trade-offs be?

This question reminds us that science does not end with explanation — it opens the door to design, innovation, and engineering. The cracked bottle is not just a physics lesson. It is an invitation to think like an engineer.

and allowing aquatic life to survive in cold conditions. Lake Victoria and other Kenyan freshwater lakes are governed by similar density-stratification principles. – Real-world engineering applications: (a) Bimetallic strip in thermostats and fire alarms — two metals with different α values bonded together; when temperature changes, unequal expansion causes the strip to bend, opening or closing an electrical circuit. (b) Expansion joints in roads, bridges (e.g., Nyali Bridge, Mombasa), and railway tracks (SGR). (c) Shrink-fitting in engineering: a metal ring is heated so it expands, slipped over a shaft, then cooled so it contracts and grips tightly — used in manufacturing. (d) Clinical (medical) thermometers relying on liquid expansion. – New driving question example: 'If engineers can calculate exactly how much a material will expand, can they design a bottle that could survive the freezing of water inside it — and what material would they use?' Connection to Phenomenon: The cracked bottle is no longer a mystery. It is a window into the particle-level behaviour of water, the anomalous physics that makes life on Earth	

possible, and the engineering principles that keep Kenya's infrastructure safe and functional.	